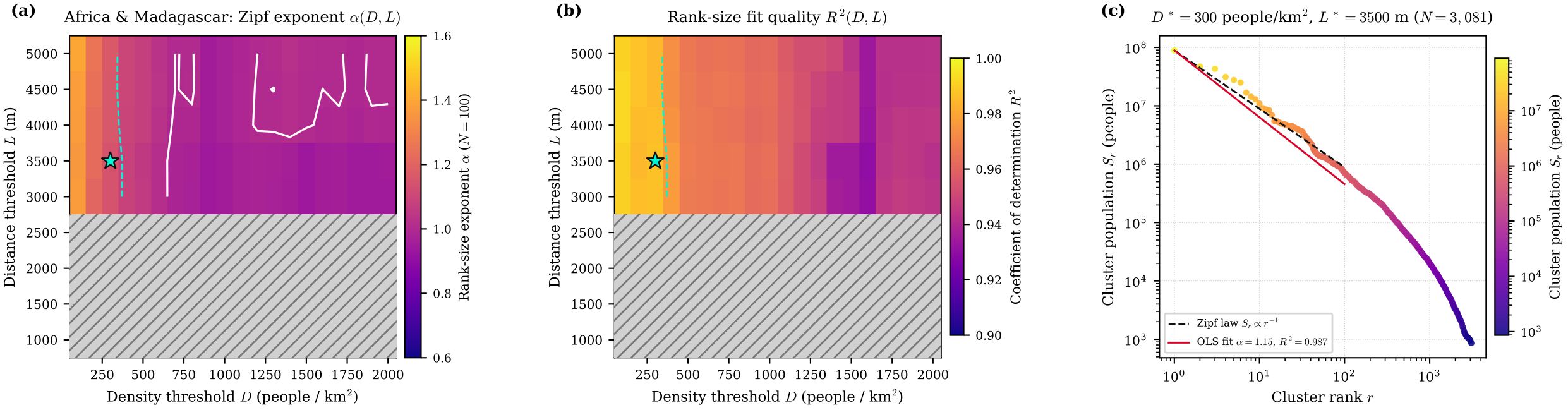

(a)
Africa & Madagascar: Zipf exponent $\alpha(D, L)$
Distance threshold L (m)
Density threshold D (people / km^2)
Rank-size exponent α ($N = 100$)
(b)
Rank-size fit quality $R^2(D, L)$
Distance threshold L (m)
Density threshold D (people / km^2)
Coefficient of determination R^2
(c)
$D^* = 300$ people/km^2, $L^* = 3500$ m ($N = 3,081$)
Cluster population S_r (people)
Cluster rank r
Zipf law $S_r \propto r^{-1}$
OLS fit $\alpha = 1.15$, $R^2 = 0.987$